

TOPICAL REVIEW

Role of Clinical Trials in Implementation Science and Organization Change: An Overview of Cluster, Stepped-Wedge, and Hybrid Effectiveness-Implementation Studies

Kelly Coughlan¹ PhD; Oyebola Fasugba² PhD; Simeon Dale³ BA (Hons); Nicola Straiton⁴ PhD; Caleb Ferguson⁵ PhD; Catherine D'Este⁶ PhD; Sandy Middleton⁷ PhD

ABSTRACT: Implementation science plays a pivotal role in advancing organizational change within stroke healthcare settings by systematically evaluating and facilitating the adoption of evidence-based practices. This topical review examines the use of key trial designs relevant to implementation science research that aim to change organizational practice, highlighting their unique design features and methodological considerations. It also provides case studies of stroke research that have used each of the methods discussed demonstrating the practical applications in real-world environments. This overview aims to guide researchers and clinicians in selecting appropriate trial designs to optimize the translation of research into sustainable organizational systems and processes.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: clinical trial ■ evidence-based practice ■ implementation science ■ stroke

Delay in translation of research findings into routine clinical practice remains an unresolved global issue.¹ This delay means that proven interventions, innovations, and best practices often take ≈2 decades to reach routine clinical use, limiting their potential to prevent complications, save lives, and enhance the efficiency of care delivery. Clinician confidence in the external validity of research findings, when making treatment decisions for those who may not have met the strict criteria of controlled trial conditions of the original research, is suggested to be a contributing factor to uptake.² The complex issue of delay from research to clinical application has seen rapid growth in the field of implementation science as the study of methods and strategies to promote uptake of evidence-based practices into real-world settings.³ With the advancement of the field, implementation science approaches are increasingly being embedded within randomized controlled trials (RCTs) to improve uptake and real-world applicability of evidence.

The core principle of implementation science is that behavior change is required at the individual, organizational or system level for successful evidence and knowledge translation. Understanding how to change behavior has led to substantial growth in the use of conceptual frameworks, models and theories, now considered best practice to guide implementation processes.^{4,5} The use of these theoretical concepts deepens understanding of the mechanisms underpinning how and why interventions work in real-world settings, and how the intervention might be sustainably embedded into routine practice. This is particularly important for complex health system interventions, which involve changing professional and organizational behaviors.⁶ Initially, these concepts were adapted from other fields such as psychology, sociology, marketing and education (eg, the Diffusion of Innovation,⁷ Social Cognitive Theory).⁸ However, there are now up to 150 conceptual frameworks, models, and theories, with many evolving from within the implementation science

Correspondence to: Sandy Middleton, PhD, Nursing Research Institute, St Vincent's Health Network Sydney, St Vincent's Hospital Melbourne & Australian Catholic University, St. Vincent's Hospital, Level 5, deLacy Bldg, 390 Victoria St, Darlinghurst 2010, New South Wales, Australia. Email sandy.middleton@acu.edu.au
Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.125.052073>.

© 2026 American Heart Association, Inc.

Stroke is available at www.ahajournals.org/journal/str

field itself. Michie et al⁹ synthesized 19 frameworks into one behavior analysis model (Capability, Opportunity, Motivation-Behavior), which links to 93 behavior change interventions (Behavior Change Wheel), the Consolidated Framework for Implementation Research,¹⁰ and the Reach, Effectiveness, Adoption, Implementation, Maintenance¹¹ are other commonly used frameworks that help researchers identify key study determinants and evaluate interventions. The choice of framework, model or theory should be matched to the implementation phase (eg, early versus sustainment or scale-up phases), previous use of these concepts in similar studies or settings and understanding the purpose of their use.⁶ Several taxonomies have also been established to enable consistency when reporting scientific methods specific to the many implementation strategies and interventions now in use. The Cochrane Effective Practice and Organisation of Care taxonomy was developed at the macro level to clarify implementation strategies specific to health-care systems (eg, delivery or financial arrangements).¹² The Expert Recommendations for Implementing Change taxonomy was created at the microlevel, to catalogue concepts specific to more discrete implementation strategies (eg, audit and feedback, reminders).¹³

The association between the number of evidence-based care processes and interventions a patient receives in hospital and improved patient outcomes is increasingly supported by research.^{14–16} Implementing these practices requires systematic and evidence-based methodological approaches. Consequently, the application of implementation science has become increasingly prominent in clinical stroke research, especially where organizational system changes are required (eg, equitable access to acute stroke unit care,¹⁷ and hyperacute stroke therapies^{18–21}).

The aim of this article is to provide an overview of 3 RCT study designs frequently used in implementation science stroke trials to evaluate interventions to facilitate behavior change at the organizational level (ie, ward level or hospital level). An outline of Cluster RCTs, Stepped-Wedge RCTs, and Hybrid Effectiveness-Implementation studies is presented with discussion of methodological features, strengths and limitations and an illustrative stroke trial case example.

This article is part of a series of articles on the Changing Landscape of Randomized Clinical Trials in Stroke. The first set of publications featured innovations in contemporary trial designs and analysis to improve efficiency, and consisted of an overview of Adaptive Clinical,²² Bayesian,²³ Adaptive Platform,²⁴ and Noninferiority²⁵ Trials. The second series of publications features innovations in stroke trial designs and evaluation that are focused on real-world implementation and clinical application. The first article of this second series provided a systematic review of study designs used to evaluate interventions aimed at changing behavior of individuals

who had experienced a stroke (ie, lifestyle modifications) to lower the risk of recurrent stroke events.²⁶

Although the study designs featured in earlier publications in this series are also applicable to implementation science research, this complementary topical review is focused on the evaluation of organizational-level interventions within stroke healthcare settings.

CLUSTER RCTs

Overview

In cluster RCTs, the unit of randomization is the cluster, so all individuals or patients within the same cluster receive the same intervention. Cluster RCTs are considered an optimal study design for testing complex implementation interventions at an organizational level and on a large scale, particularly in health service and global health research.^{27–29} In cluster RCTs, preexisting, natural or self-selected groups—clusters, such as hospitals or wards—are randomized. This approach is used when individual randomization is not feasible, when there is a potential for treatment contamination between participants exposed to the intervention and those who are not, or when the intervention is aimed at the organizational unit (ie, the cluster level).³⁰ Clustering can occur at various and multiple levels, such as patients clustered within ward/unit/department or hospitals within a healthcare organization.^{31,32} Although randomization occurs at the cluster level, outcomes are generally measured at the individual level.^{28,32}

This study design can be used to assess changes to group behaviors, leading to large-scale effects.²⁸ This approach is also particularly suited to interventions targeting healthcare professionals and organizational behavior change.²⁹ For example, implementing stroke care protocols requires changes to stroke care pathways and workflows, which also require leadership engagement and multidisciplinary training. These elements operate at the organizational level, requiring randomization at the organizational, not the individual patient level, to minimize contamination of the intervention. In implementation science cluster RCTs, the clinical component of the intervention (eg, a care protocol) is the evidence-based best practice, and it is the implementation strategy component of the intervention that is being tested (ie, how to achieve clinician behavior change, for example, education, reminders). Due to the level of organizational complexity and possibly large changes to clinical practices, interventions usually require a bedding-down phase before collection of postimplementation data. After cluster randomization, this transition phase allows for routines and workflow to normalize before formal postimplementation outcome assessment. This approach acknowledges that intervention effects may not be immediate and that fidelity and process measures are prioritized, primary analyses will, therefore, exclude data accrued during this period.³³ No

ideal length of time has been identified but 3 months has been reported in some trials as adequate for the new practice to become business as usual.^{34,35}

METHODOLOGICAL FEATURES OF CLUSTER RCTs

Cluster RCTs can take various forms. In a simple cluster RCT, clusters are randomized, and each cluster remains in its assigned intervention arm for the duration of the study (Figure [A]). Outcomes can be assessed at baseline as well as follow-up, and baseline outcomes adjusted for in the analysis.³⁶ The cross-over cluster RCT is an alternative, in which all clusters receive the intervention (Figure [B]). Clusters are initially randomized to one intervention arm, and after a defined period, they are transitioned to the other arm. However, this design introduces the risk of carryover effects potentially leading to contamination across intervention arms.³⁷ In some crossover cluster RCTs, a prespecified washout period is used between the end of one intervention phase and the start of the next, allowing any residual effects of the first intervention to dissipate before the second begins.³⁸ Another option is a partial crossover cluster RCT, which can reduce the risk of contamination of a standard crossover design (Figure [C]). Using this method, data are collected at baseline across all clusters, followed by intervention delivery to half the clusters during the next phase. Although this mitigates contamination risks, not all clusters receive the intervention.³⁷ The stepped-wedge design is a staggered cluster trial design where the sequence of transition from control to intervention is randomized for all clusters (Figure [D]).³⁷ This method is discussed in more detail in the next section.

In cluster RCTs, outcomes for individuals within a cluster are generally correlated. Individuals within clusters are more similar than individuals from different clusters; that is, there is greater homogeneity within than between clusters in terms of measured and unmeasured covariates.³⁹ This has implications for

statistical power and analysis, as standard methods for power calculation and statistical analysis assume independent observations. The correlation within clusters is usually quantified by the intracluster correlation coefficient, which indicates the proportion of the variance in outcome values that is explained by the variability between clusters. It is commonly expressed as the ratio of the between-cluster variance and the total variance.⁴⁰ Appropriate methods of sample size and power estimation need to account for the correlation of outcomes within clusters, generally resulting in larger numbers than individually randomized trials. In cluster RCTs, the design effect, which is driven by the intracluster correlation coefficient inflates the variance and therefore increases the required sample size. Importantly, statistical power is typically driven not only by the number of participants per cluster but to a greater extent by the number of clusters, highlighting the importance of prioritizing cluster recruitment over enlarging the cluster size.^{32,41} Variability in cluster size can also reduce statistical power, and should be considered in power calculations.^{42,43} Many statistical packages can estimate sample size for cluster RCTs, using information on the number of clusters, number of participants per cluster and the likely correlation within or between clusters.^{44–46}

Using standard methods (which assume independence) for correlated data can underestimate the variance, resulting in the incorrect interpretation of coefficients (CIs too narrow and *P* values too small), in addition to generating potentially biased estimates for intervention effects, and lower than expected statistical power.³⁸

Appropriate methods for analysis of cluster RCTs require weighted estimates of parameters with SEs which are adjusted for correlation of outcomes; the most commonly used approaches are mixed models (also known as random effects models, multilevel models, hierarchical models)⁴⁷ and Generalized Estimating Equations (GEEs).⁴² In mixed models, the assumption is that the intervention effect varies between clusters, and the overall effect estimate is a weighted average of the cluster-specific effects. Thus, mixed models have

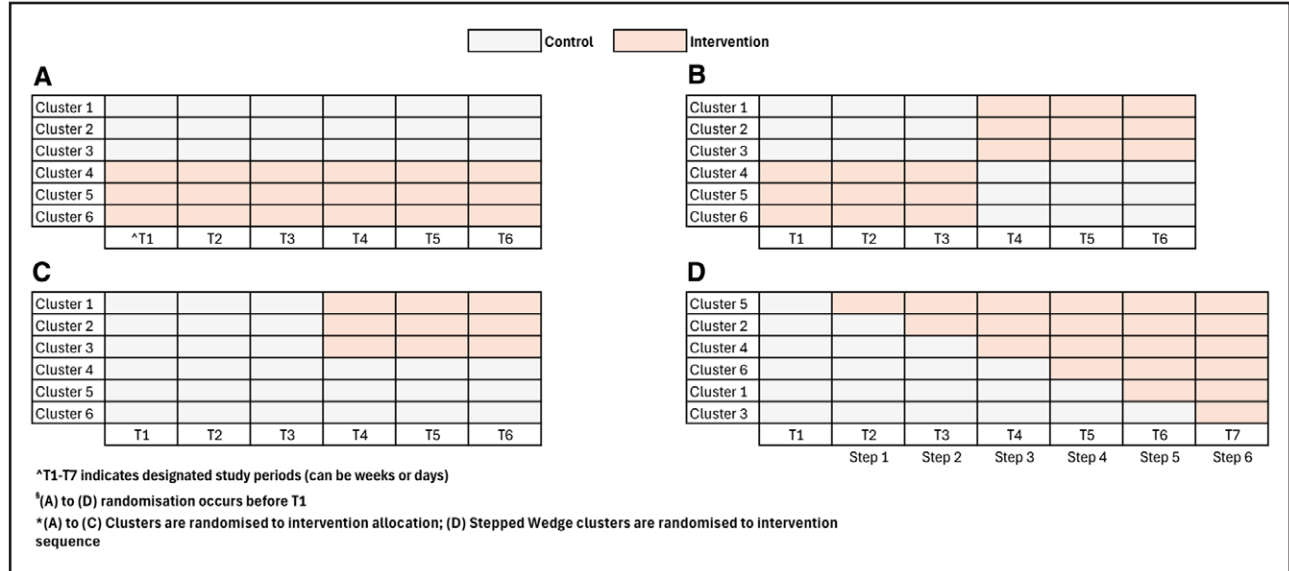


Figure. Intervention allocation and sequencing of cluster randomized controlled trials. ^T1–T7 indicates designated study periods (can be weeks or days). **A** through **D**, Randomization occurs before T1. **A** through **C**, Clusters are randomized to intervention allocation; **(D)** stepped-wedge clusters are randomized to intervention sequence.

a cluster-specific interpretation. A GEE approach to analysis of correlated data assumes that the intervention effect is the same for all clusters, what is known as a marginal or population-average approach.⁴³

The choice of approach may depend on various aspects of the study design, including the research question, type of outcome, distributional assumptions, and the number of clusters available. Mixed-effects models estimate effects for specific groups (subjects, hospitals), accounting for their unique variation under the assumption of normally distributed random effects.⁴⁷ If outcomes are specific to individual patients in a particular hospital, a mixed-effects model should be used. GEEs estimate effects for the entire population, adjusting for clustering to ensure correct SEs but without modeling the individual group differences.⁴² If outcomes are aimed at estimating a population-average effect for patients in a target population, then GEE should be used.⁴⁰

The number of clusters in a cluster RCT can be small and this can result in baseline imbalances in covariates. Randomization of clusters can be stratified by factors known to be highly correlated with outcome, to minimize imbalance.³² Another potential strategy is to use baseline values of the outcome in the analysis. This can involve using a cluster-level summary value of the outcome as a covariate in the regression model.⁴⁰ Alternatively, in cluster RCTs that collect data over several timepoints, individual values of the outcome can be included in a regression model and a time by intervention group interaction term used to determine if a change from baseline to follow-up differs between intervention and control groups.³²

Given the complexity of the methodological features and considerations in relation to cluster RCTs, the Consolidated Standards of Reporting Trials (CONSORT) statement used to guide the transparent reporting of RCTs, has been updated with an extension specific to cluster RCTs. The additional reporting requirements consist of the provision of a strong rationale for the choice of study design, how the effects of clustering have been incorporated into sample size calculations and analysis, and the flow of both clusters and individuals through the trial.⁴⁸

Strengths and Limitations of Cluster RCTs

The key strength of cluster RCTs is their ability to minimize contamination between intervention and control groups, particularly when individuals within the same cluster (ie, ward or hospital) can potentially influence each other's behaviors or outcomes. This design is also logistically and ethically advantageous when interventions are delivered at the group level, when individual randomization is impractical or inappropriate, or when one is testing the implementation of a practice of known benefit.⁴⁹

Despite their strengths, cluster RCTs are known for their methodological challenges (eg, a larger sample size is required for adequate statistical power to detect an intervention effect); and the complexity of statistical analysis and reporting as discussed in the previous section.³² Other limitations consist of the risk of selection bias as clusters are generally selected before randomization, which may impact on generalizability. To minimize this risk, recruitment and randomization should be

undertaken by 2 separate individuals and not by the person delivering the intervention.^{28,32} Maintaining allocation concealment may be difficult³⁷ and blinding of clinicians delivering the intervention is often not feasible (although blinding of outcome assessment should still be possible).⁴⁸ Cluster RCTs also raise ethical concerns around informed consent.³¹ There are several key references to guide navigation around the methodological considerations specific to cluster RCTs.^{32,42,43} See [Box S1](#) for an illustrative stroke study example of a cluster RCT (the QASC trial [Quality in Acute Stroke Care]).⁵⁰

STEPPED-WEDGE CLUSTER RCTs

Overview

The SW-CRCT (Stepped-Wedge Cluster Randomized Trial) has become increasingly popular, particularly in health services and health systems-related research. It is useful for the evaluation of service-level interventions (such as implementation of best practice protocolized care), allowing all participating clusters to receive the interventions and supporting data collection across different stages of the roll-out. The design involves sequential movement of clusters from control to intervention status in a randomized sequence, until all clusters have received the intervention. Each transition from control to intervention is a step in the study (Figure [D]). Unlike a standard crossover trial, the SW-CRCT transition is from control to intervention only, so it is considered to be a 1-way crossover cluster RCT.⁵¹ Historically, they have also been referred to as phased implementation studies.

A SW-CRCT is generally favored over a standard cluster RCT when the number of clusters available is small, as in some circumstances a SW-CRCT can be a more efficient design (ie, require fewer clusters than a standard cluster RCT), or when it is a requirement, for ethical or practical reasons, for all clusters to receive the intervention.⁵¹ In implementation trials, a SW-CRCT also can be used where there is existing evidence of the benefit of the clinical component of the intervention, and only the effectiveness of the implementation strategy is being tested (ie, how to achieve clinician behavior change). The outcome measure in such trials would reflect the success of the implementation strategy and could include measures of behavior change, such as clinician adoption, fidelity to the intervention or reach.⁵²

METHODOLOGICAL FEATURES OF STEPPED-WEDGE CLUSTER RCTs

In SW-CRCTs, clusters are randomized to transition from control to intervention phase either one at a time or in groups. The number of clusters randomized to transition does not need to be consistent at each step. The outcome is measured on all clusters at multiple points in time: at baseline, while all clusters

are in the control phase, and then after each step, when one or more clusters move from control to intervention. Follow-up assessments are generally also conducted at a common end point after all clusters have completed transition to the intervention.^{36,51} Multiple outcome measures over time for each cluster can provide a gain in statistical efficiency.⁵³

SW-CRCTs have the same methodological issues as cluster RCTs, but with the added complication that, because all clusters begin in the control phase and end in the intervention phase, there is a potential time trend which must be accounted for.⁵⁴ In addition, there are multiple measures of the outcome across time. The time trend is adjusted for in the analysis by including a time variable (continuous or categorical) in the analysis model, as well as outcomes at each time and the intervention phase of each cluster at each step. Adjustment for the trend in outcomes over time can reduce the statistical efficiency or precision of the study, while multiple measures of the outcome over time can potentially increase precision as each cluster provides outcomes in both control and intervention phases.^{54,55}

The multiple measures of the outcome and adjustment for time trend must be considered when estimating sample size for a SW-CRCT. The efficiency of a SW-CRT relative to a standard cluster RCT depends on the intraclass correlation and the cluster size of the study.^{54–56} SW-CRCTs are sometimes favored over standard cluster RCTs as they can achieve equivalent power with fewer clusters; however, the relative statistical efficiency of these 2 designs depends on a range of factors including number of clusters, number of data collection points, and the number of participants per cluster.⁵³ There is also an extension to the CONSORT statement specific to SW-CRCTs to guide researchers on their reporting of these complex methodological features to improve transparency and interpretability of findings.⁵⁷

Strengths and Limitations of Stepped-Wedge Cluster RCT Design

A key strength of this design is that it allows for ethical or equitable access to the intervention, ensuring all clusters receive the intervention. This can be helpful in the delivery of health system-level interventions, such as changes to systems or models of care. The pragmatic nature of the design can help navigate some logistical challenges or constraints experienced with more traditional study designs. Patient-level consent to the intervention itself and, hence, individual patient recruitment may not be required, as sites are randomized at the group level, rather than individually. Often, service-level consent or an opt-out or waiver of consent may be used as the clinical component of the intervention is evidence-based and already known to be effective.

A key limitation for SW-CRCTs and all multicenter studies relates to the cost of conducting the study and the challenging logistics. As the number of clusters increases, so too does the administrative burden of ethics and governance applications and financial requirements. Time-related confounding, cluster-level variability,⁵⁶ and the complexity of statistical analyses (applicable to all types of cluster RCTs) are additional potential challenges of this study design.

See [Box S2](#) for an illustrative stroke study example of a Stepped-Wedge Cluster RCT (the OPTIMISTmain Trial [Optimal Post tPA IV Monitoring in Ischaemic Stroke]).⁵⁸

HYBRID EFFECTIVENESS-IMPLEMENTATION STUDIES

Overview

The conventional research pathway for translating evidence-based clinical interventions into routine practice typically follows a sequential, staged process.⁵⁹ This process prioritizes proving an intervention's efficacy under controlled conditions, followed by effectiveness trials and systematic reviews, before addressing its adoption in real-world settings. Although rigorous, this approach often results in significant delays between developing an intervention and achieving widespread implementation in clinical practice.⁵⁹ To address this gap, Hybrid Effectiveness-Implementation studies have been introduced to streamline the process and accelerate the integration of evidence-based interventions into standard care.⁶⁰ The use of Hybrid Effectiveness-Implementation studies is also an acknowledgement that new organizational level interventions are required to facilitate clinician behavior change for the successful implementation of effective patient management and care delivery strategies. To date, the majority of Hybrid Effectiveness-Implementation studies have been RCTs; however, other nonrandomized methods are now being included in these designs.⁶⁰

Hybrid Effectiveness-Implementation Trials, of which there are 3 main types, simultaneously assess the effectiveness of an intervention and examine implementation outcomes, processes, strategies, and contextual factors influencing its adoption and sustainability in practice, often within a single trial.⁶⁰ In a type I Hybrid Effectiveness-Implementation Trial, the primary aim is to test the clinical effectiveness of an intervention (may be new or untested in the target population) with collection of some secondary implementation measures. A type II Hybrid Effectiveness-Implementation Trial assigns equal importance to both the clinical effectiveness and implementation outcomes (ie, coprimary outcomes). Conversely, a type III Hybrid Effectiveness-Implementation Trial evaluates implementation outcomes as the primary focus and effectiveness outcomes as secondary (eg, clinical intervention is evidence-based but under-used, the implementation strategy/s is being tested to evaluate whether it results in practice change⁶⁰; Table).

METHODOLOGICAL FEATURES OF HYBRID EFFECTIVENESS-IMPLEMENTATION TRIALS

Hybrid Effectiveness-Implementation Trials can be individually or cluster randomized.⁶² The level of randomization depends

Table. Overview of Hybrid Effectiveness-Implementation Study Types

Hybrid type	Primary aim/outcome	Secondary aim/outcome	Example research question
Type 1	Effectiveness of the clinical intervention on relevant effectiveness outcomes (eg, clinical end points, behavior change)	Collects limited data on implementation processes/outcomes (eg, adoption, acceptability, fidelity)	Does the intervention improve patient outcomes in real-world settings?
Type 2	Equal emphasis on intervention effectiveness and implementation outcomes (eg, coprimary outcomes)	Not applicable (both given equal weight)	Does the intervention improve outcomes and is the chosen implementation strategy effective?
Type 3	Testing the implementation strategy on relevant implementation outcomes (eg, adoption, acceptability, fidelity)	Collects limited data on effectiveness outcomes (eg, clinical end points, behavior change)	Is the implementation strategy effective in increasing uptake of the clinical intervention?

Informed by Curran et al.⁶¹

on the intervention and context, but in general, type I Hybrid Effectiveness-Implementation Trials often randomize individual patients and observe implementation, whereas type II and type III Hybrid Effectiveness-Implementation Trials may randomize clusters to test both intervention and implementation strategies simultaneously.⁶⁰

Separate analyses are typically conducted for effectiveness and implementation outcomes. Multilevel or GEE models are often used to account for clustering and time effects in primary analysis. Mixed methods are generally used for addressing the secondary aims.⁶³ Depending on the specific trial aims, this can involve formally or descriptively comparing the specified secondary measures between groups (quantitative analysis), followed by qualitative analysis, such as semistructured interviews or focus groups, to explore stakeholder perspectives, contextual variables, and mechanisms of change.⁶⁴

The study design is dictated by the research question and outcomes of interest. Type I Hybrid Effectiveness-Implementation Trials are powered to investigate the effectiveness outcome/outcomes (ie, clinical end points/behavior change). There is generally one primary, with some secondary, effectiveness outcome/s. Data on implementation measures (eg, adoption, acceptability, fidelity) can be reported descriptively, or considered as secondary or tertiary level outcomes, depending on the aims of the study (eg, Does the intervention improve patient outcomes in real-world settings?). Type II Hybrid Effectiveness-Implementation Trials give equal weight to effectiveness and implementation, require careful planning as there are 2 coprimary end points (one effectiveness outcome and one implementation outcome), and the research design options are vast (eg, Does the intervention improve outcomes and is the chosen implementation strategy effective?).

Having >1 primary outcome introduces multiple testing and increases the risk of Type I errors (when the null hypothesis is incorrectly rejected). The implications of this need to be considered in the context of the hypothesis testing framework chosen to guide the power analysis.⁶² A recent literature review of Cluster Randomized Hybrid Type II Effectiveness-Implementation Studies reported there were 5 key statistical methods for handling multiple primary outcomes: *P* value adjustments, combined outcomes approach, single 1 degree of freedom combined test for multiple outcomes, disjunctive 2 degree of freedom test for multiple outcomes, and the conjunctive test.⁶³ Further elaboration on these statistical methods is beyond the scope of this publication; however, researchers must be aware of these complexities when undertaking or considering this study design, especially in the context of limited statistical software to support methods for analysis of

coprimary outcomes.⁶⁴ In the absence of a CONSORT extension specific to Hybrid Effectiveness-Implementation Trials, the Standards for Reporting Implementation Studies guideline is increasingly used to guide comprehensive reporting of both the clinical intervention and the implementation strategy.⁶⁵

Acknowledging the dual focus of effectiveness-implementation trials emphasizes the importance of having both an implementation scientist and a biostatistician with experience in these methods as essential members of the research team. The implementation frameworks by Proctor et al⁵² and Reach, Effectiveness, Adoption, Implementation, Maintenance¹¹ are widely used in type II Hybrid Effectiveness-Implementation Trials to define implementation outcomes that can be measured quantitatively.⁶² Commonly, implementation outcomes consist of: Acceptability, Adoption, Appropriateness, Feasibility, Fidelity, Penetration, Sustainability and Implementation costs⁶² as determined by the theoretical framework guiding the study design.

Type III Hybrid Effectiveness-Implementation Trials are often powered for evaluation of quantitative implementation outcomes as the primary (and potentially secondary) measure; that is, when the evidence-base or effectiveness of the intervention is considered to be established, and the primary goal is to understand how the implementation strategy or strategies affect implementation outcomes⁶⁰ (eg, Is the implementation strategy effective in increasing uptake of the clinical intervention?). Effectiveness measures can be considered descriptively as process measures, or as secondary outcomes, or both, as determined by the individual study aims.⁶⁰

Strengths and Limitations of Hybrid Effectiveness-Implementation Trials

Hybrid Effectiveness-Implementation Trial designs enhance resource efficiency by reducing the need for sequential research, thereby saving both time and costs.^{66,67} Their adaptability allows alignment with pragmatic trial methods to improve external validity. In addition, the use of mixed methods (eg, quantitative outcome evaluation and qualitative methods in a process evaluation) provides key context for interpreting results,⁶⁸ understanding intervention fidelity and informing potential scale-up to other populations.

However, given the relatively recent development and application of Hybrid Effectiveness-Implementation randomized and nonrandomized studies, there remain several complexities that require further consideration

and refinement. In addition to the careful planning to appropriately measure and power both effectiveness and implementation outcomes, data collection is often resource-intensive, spanning multiple levels including patients, providers, and organizations. Balancing focus between clinical and implementation outcomes can lead to potential trade-offs, sometimes diluting statistical power for one or both measures.⁶⁸ In addition, interpreting results can be challenging, especially when effectiveness and implementation findings diverge, making it difficult to draw clear conclusions. Successful execution of all Hybrid Effectiveness-Implementation Trials requires multidisciplinary expertise, necessitating close collaboration between clinical researchers, implementation scientists, statisticians, and consumers. See [Box S3](#) for an illustrative stroke study example of type II Hybrid Effectiveness-Implementation design (TCSI trial [Transitional Care Stroke Intervention])^{69,70}

DISCUSSION

This overview summarizes study designs frequently used in implementation science stroke trials to evaluate interventions to facilitate behavior change at the organizational level. The landscape of clinical trials in stroke is likely to continue to see an increase in the use of implementation science in the quest to rapidly translate research findings into everyday clinical practice. Bridging this gulf and reducing the unacceptable time delay from bench to bedside is a key priority for researchers and policy makers globally. The conceptual theories, models and frameworks that now exist to guide this evolving field must be harnessed to strengthen study designs and interpret study findings appropriately. The additional challenges of implementation science trials that relate to increased statistical complexity and methodological intricacies can be offset with careful study planning and the inclusion of implementation experts and biostatisticians within study teams.

A strength of implementation science trials is their pragmatic nature, that is, their ability to evaluate interventions under usual or standard care conditions, thereby increasing the generalizability and applicability of trial findings.³¹ Hence, interventions can be applied as they would be in the real world as opposed to the tightly controlled conditions of traditional RCTs.³⁷ This enhanced external validity is particularly evident when interventions involve large clusters, such as health services. Furthermore, evidence from cluster RCTs is particularly attractive to policymakers, as they more closely reflect implementation in clinical practice, supporting broader system change.²⁸

However, achieving study end points can be considerably harder to demonstrate in pragmatic real-life health systems, given the influence of widely variable contextual professional and organizational factors.⁷¹ Improving our

understanding of how and why interventions fail or succeed in different populations is critical to wider uptake of evidence-based practices and the sustainability of strategies in the longer term.⁷² The Pragmatic-Explanatory Continuum Indicator Summary 2 tool is increasingly used to assist researchers at the design stage to ensure their chosen strategies for implementation and evaluation accurately reflect the intended aims of the study.⁷³

Ensuring intervention fidelity is a key concern for all clinical trials, particularly when the intervention is being delivered as part of usual or standard care processes by the stroke multidisciplinary team. It is important to consider how sites will be trained to provide the intervention in a consistent approach and how intervention fidelity will be managed. Special attention should be paid to ensure that although contextual adaptations are expected, the core or minimal mandatory care elements of the intervention should not change over time or across sites, and the implementation strategy remains consistent.⁷⁴

The use of a conceptual framework, model or theory is recommended best practice for implementation research. Their use strengthens the study design and can inform process evaluation studies (conducted during or after the intervention trials to understand if intervention fidelity was maintained and its interaction with context).⁷⁵ Choosing the most appropriate framework, model or theory in the setting of >150 options, however, can be overwhelming. The Theory, Model, and Framework Comparison and Selection Tool⁷⁶ and the Systematic Evaluation and Selection of Implementation Science Theories, Models and Frameworks⁷⁷ meta-framework are excellent resources to guide such decisions. The inclusion of an implementation expert as a coinvestigator with experience in the use of these frameworks is essential for the effective execution of the study design.

Additional challenges in implementation science research are the reporting of these complex trial designs. Despite the existence of taxonomies for consistency when describing implementation strategies, and the many frameworks, models and theories to guide their use, there is often insufficient detail on the strategies and target behaviors to enable replication of methods for future research. A recent review of 204 implementation science trials found that tailoring, modification, and fidelity were often underreported. These concepts are essential to understanding the effectiveness of implementation strategies and relate directly to their sustainability in clinical practice.⁷⁸ Integration of a reporting framework (eg, CONSORT and CONSORT extension/s, Template for Intervention Description and Replication checklist) and a behavioral change framework (eg, Action, Actor, Context, Target, Time) in the planning and reporting study phases is recommended to strengthen research rigour and continue to advance the field of implementation science.⁷⁸ The importance of detailed reporting to support study replication in other populations is critical to scaling

up effectively, which is directly in line with the broader global directive of health equity. The Lancet Commission on Evidence-Based Implementation in Global Health⁷⁹ was recently established to strengthen the use of implementation science in global health, especially in low- and middle-income countries, through the use of context-specific strategies that support equitable, scalable, and sustainable health interventions.

Recognition of the transformative potential of a systematic and theory-informed approach to achieving organizational change is growing, particularly in complex clinical areas like stroke. The field of implementation science is poised for continued growth and applicability in stroke care trials, propelled by innovation, collaboration, and a shared commitment to improving patient outcomes and enhancing equitable care delivery.

ARTICLE INFORMATION

Affiliations

Nursing Research Institute—St Vincent's Health Network Sydney & St Vincent's Hospital Melbourne, Australia (K.C., O.F., S.D., N.S., S.M.). School of Nursing, Midwifery and Paramedicine, North Sydney, Australian Catholic University, Australia (K.C., O.F., S.D., N.S., S.M.). Centre for Chronic & Complex Care Research, Blacktown Hospital, Western Sydney Local Health District, Australia (C.F.). School of Nursing, University of Wollongong, Australia (C.F.). The Sax Institute, Sydney, Australia (C.D.). School of Medicine and Public Health, The University of Newcastle, Australia (C.D.).

Sources of Funding

None.

Disclosures

Dr Ferguson reports compensation from National Stroke Foundation for other services; compensation from Cardiac Society of Australia and New Zealand for other services; compensation from National Stroke Foundation for other services; employment by National Health and Medical Research Council; and employment by Heart Foundation. Dr Middleton receives support from a National Health and Medical Research Council Investigator Grant (APP1196352). The other authors report no conflicts.

Supplemental Material

Case Studies (QASC Trial, OPTIMISTMain Trial, TCSI Trial)

Boxes S1–S3

References 10,25,50,58,69,70,73,74,80–95

REFERENCES

- Braithwaite J, Glasziou P, Westbrook J. The three numbers you need to know about healthcare: the 60-30-10 challenge. *BMC Med*. 2020;18:102. doi: 10.1186/s12916-020-01563-4
- Rothwell PM. External validity of randomised controlled trials: "To whom do the results of this trial apply?" *Lancet*. 2005;365:82–93. doi: 10.1016/S0140-6736(04)17670-8
- Chambers DA, Emmons KM. Navigating the field of implementation science towards maturity: challenges and opportunities. *Implement Sci*. 2024;19:26. doi: 10.1186/s13012-024-01352-0
- Atkins L, Francis J, Islam R, O'Connor D, Patey A, Ivers N, Foy R, Duncan EM, Colquhoun H, Grimshaw JM, et al. A guide to using the Theoretical Domains Framework of behaviour change to investigate implementation problems. *Implement Sci*. 2017;12:77. doi: 10.1186/s13012-017-0605-9
- Lynch EA, Mudge A, Knowles S, Kitson AL, Hunter SC, Harvey G. "There is nothing so practical as a good theory": a pragmatic guide for selecting theoretical approaches for implementation projects. *BMC Health Serv Res*. 2018;18:857. doi: 10.1186/s12913-018-3671-z
- Nilsen P. Making sense of implementation theories, models and frameworks. *Implement Sci*. 2015;10:53. doi: 10.1186/s13012-015-0242-0
- Rogers EM. A Prospective and retrospective look at the diffusion model. *J Health Commun*. 2004;9(suppl 1):13–19. doi: 10.1080/10810730490271449
- Van Lange PAM, Kruglanski AW, Higgins ET. Social cognitive theory. In: Van Lange PAM, Kruglanski AW, Higgins ET, eds. *Handbook of Theories of Social Psychology: Volume 1*. Sage Publications; 2012.
- Michie S, van Stralen MM, West R. The behaviour change wheel: a new method for characterising and designing behaviour change interventions. *Implement Sci*. 2011;6:42. doi: 10.1186/1748-5908-6-42
- Damschroder LJ, Aron DC, Keith RE, Kirsh SR, Alexander JA, Lowery JC. Fostering implementation of health services research findings into practice: a consolidated framework for advancing implementation science. *Implement Sci*. 2009;4:50. doi: 10.1186/1748-5908-4-50
- Glasgow RE, Vogt TM, Boles SM. Evaluating the public health impact of health promotion interventions: the RE-AIM framework. *Am J Public Health*. 1999;89:1322–1327. doi: 10.2105/ajph.89.9.1322
- Effective Practice and Organisation of Care (EPOC). EPOC taxonomy. 2021. Accessed November 5, 2025. <https://doi.org/10.5281/zenodo.5105851>
- Powell BJ, Waltz TJ, Chinman MJ, Damschroder LJ, Smith JL, Matthieu MM, Proctor EK, Kirchner JE. A refined compilation of implementation strategies: results from the Expert Recommendations for Implementing Change (ERIC) project. *Implement Sci*. 2015;10:21. doi: 10.1186/s13012-015-0209-1
- Cadilhac DA, Andrew NE, Lannin NA, Middleton S, Levi CR, Dewey HM, Grabsch B, Faux S, Hill K, Grimley R, et al; Australian Stroke Clinical Registry Consortium. Quality of acute care and long-term quality of life and survival: the Australian Stroke Clinical Registry. *Stroke*. 2017;48:1026–1032. doi: 10.1161/STROKEAHA.116.015714
- Connor L, Dean J, McNett M, Tydings DM, Shrout A, Gorsuch PF, Hole A, Moore L, Brown R, Melnyk BM, et al. Evidence-based practice improves patient outcomes and healthcare system return on investment: findings from a scoping review. *Worldviews Evid Based Nurs*. 2023;20:6–15. doi: 10.1111/wvn.12621
- Song S, Fonarow GC, Olson DM, Liang L, Schulte PJ, Hernandez AF, Peterson ED, Reeves MJ, Smith EE, Schwamm LH, et al. Association of Get With The Guidelines-Stroke program participation and clinical outcomes for Medicare beneficiaries with ischemic stroke. *Stroke*. 2016;47:1294–1302. doi: 10.1161/STROKEAHA.115.011874
- Langhorne P, Ramachandra S; Stroke Unit Trialists' Collaboration. Organised inpatient (stroke unit) care for stroke: network meta-analysis. *Cochrane Database Syst Rev*. 2020;4:CD000197. doi: 10.1002/14651858.CD000197.pub4
- Wardlaw JM, Murray V, Berge E, del Zoppo GJ. Thrombolysis for acute ischaemic stroke. *Cochrane Database Syst Rev*. 2014;7:CD000213. doi: 10.1002/14651858.CD000213.pub3
- Huo X, Ma G, Tong X, Zhang X, Pan Y, Nguyen TN, Yuan G, Han H, Chen W, Wei M, et al; ANGEL-ASPECT Investigators. Trial of endovascular therapy for acute ischemic stroke with large infarct. *N Engl J Med*. 2023;388:1272–1283. doi: 10.1056/NEJMoa2213379
- Lazarus G, Permana AP, Nugroho SW, Audrey J, Wijaya DN, Widyahening IS. Telestroke strategies to enhance acute stroke management in rural settings: a systematic review and meta-analysis. *Brain Behav*. 2020;10:e01787. doi: 10.1002/brb3.1787
- Turc G, Hadziahmetovic M, Walter S, Churilov L, Larsen K, Grotta JC, Yamal JM, Bowry R, Katsanos AH, Zhao H, et al. Comparison of mobile stroke unit with usual care for acute ischemic stroke management: a systematic review and meta-analysis. *JAMA Neurol*. 2022;79:281–290. doi: 10.1001/jamaneurol.2021.5321
- Crawford AM, Lorenzi EC, Saville BR, Lewis RJ, Anderson CS. Adaptive clinical trials in stroke. *Stroke*. 2024;55:2731–2741. doi: 10.1161/STROKEAHA.124.046125
- Ospel JM, Brown S, Holodinsky JK, Rinkel L, Ganesh A, Coutts SB, Menon BK, Saville BR, Hill MD, Goyal M. An introduction to Bayesian approaches to trial design and statistics for stroke researchers. *Stroke*. 2024;55:2742–2753. doi: 10.1161/STROKEAHA.123.044144
- Lorenzi E, Crawford AM, Anderson CS, Menon BK, Chen X, Mistry E, Khatri P, Elm JJ, Beall J, Saville BR, et al. Adaptive platform trials in stroke. *Stroke*. 2025;56:198–208. doi: 10.1161/STROKEAHA.124.045754
- Caicalante F, Treurniet KM, Kappelhof M, Kaesmacher J, Lingsma HF, Saver JL, Gralla J, Fischer U, Majoie CB, Roos YB. Understanding noninferiority trials: what stroke specialists should know. *Stroke*. 2025;56:543–552. doi: 10.1161/strokeaha.124.048024
- Naqvi IA, Ebadi SR, Patel SR, George M, Cheung YK, Kronish IM. Intervention design for individual behavior change in stroke prevention trials. *Stroke*. 2026;57:xxx–xxx. doi: 10.1161/STROKEAHA.126.056667

27. Barratt H, Campbell M, Moore L, Zwarenstein M, Bower P. Randomised controlled trials of complex interventions and large-scale transformation of services. *BMJ Open*. 2016;16:e009768.
28. Dron L, Taljaard M, Cheung YB, Grais R, Ford N, Thorlund K, Jahan F, Nakimuli-Mpungu E, Xavier D, Bhutta ZA, et al. The role and challenges of cluster randomised trials for global health. *Lancet Glob Health*. 2021;9:e701–e710. doi: 10.1016/S2214-109X(20)30541-6
29. Grimshaw J, Eccles M, Campbell M, Elbourne D. Cluster randomized trials of professional and organizational behavior change interventions in health care settings. *ANN Am Acad Polit Soc Sci*. 2005;599:71–93. doi: 10.1177/0002716205274576
30. Foy R, Ivers NM, Grimshaw JM, Wilson PM. What is the role of randomised trials in implementation science? *Trials*. 2023;24:537. doi: 10.1186/s13063-023-07578-5
31. Wolfenden L, Foy R, Presseau J, Grimshaw JM, Ivers NM, Powell BJ, Taljaard M, Wiggers J, Sutherland R, Nathan N, et al. Designing and undertaking randomised implementation trials: guide for researchers. *BMJ*. 2021;372:m3721. doi: 10.1136/bmj.m3721
32. Hemming K, Taljaard M. Key considerations for designing, conducting and analysing a cluster randomized trial. *Int J Epidemiol*. 2023;52:1648–1658. doi: 10.1093/ije/dyad064
33. Watson SI, Hooper R. Grand rounds in methodology: improving the design of staggered implementation cluster randomised trials. *BMJ Qual Saf*. 2025;34:631–636. doi: 10.1136/bmjqs-2025-018697
34. Middleton S, Dale S, Cheung NW, Cadilhac DA, Grimshaw JM, Levi CR, McInnes E, Considine J, McElduff P, Gerraty R, et al. Nurse-initiated acute stroke care in emergency departments: the triage, treatment, and transfer implementation cluster randomized controlled trial. *Stroke*. 2019;50:1346–1355. doi: 10.1161/STROKEAHA.118.020701
35. McBain RK, Mwale O, Mpinga K, Kamwiyo M, Kayira W, Ruderman T, Connolly E, Watson SI, Wroe EB, Munyaneza F, et al. Effectiveness, cost-effectiveness, and positive externalities of integrated chronic care for adults with major depressive disorder in Malawi (IC3D): a stepped-wedge, cluster-randomised, controlled trial. *Lancet*. 2024;404:1823–1834. doi: 10.1016/S0140-6736(24)01809-9
36. Hooper R, Forbes A, Hemming K, Takeda A, Beresford L. Analysis of cluster randomised trials with an assessment of outcome at baseline. *BMJ*. 2018;360:k1121. doi: 10.1136/bmj.k1121
37. Cook AJ, DeLong ER, Murray DM, Vollmer WM, Heagerty PJ. Statistical lessons learned for designing cluster randomized pragmatic clinical trials from the NIH Health Care Systems Collaboratory Biostatistics and Design Core. *Clin Trials*. 2016;13:504–512. doi: 10.1177/1740774516646578
38. Shi D, Ye T. Behavioral carry-over effect and power consideration in cross-over trials. *Biometrics*. 2024;80:ujae023. doi: 10.1093/biomet/ujae023
39. Giraudeau B, Weijer C, Eldridge SM, Hemming K, Taljaard M. Why and when should we cluster randomize? *J Epidemiol Popul Health*. 2024;72:202197. doi: 10.1016/j.jep.2024.202197
40. Leyrat C, Morgan KE, Leurent B, Kahan BC. Cluster randomized trials with a small number of clusters: which analyses should be used? *Int J Epidemiol*. 2017;47:321–331. doi: 10.1093/ije/dyx169
41. Leyrat C, Eldridge S, Taljaard M, Hemming K. Practical considerations for sample size calculation for cluster randomized trials. *J Epidemiol Popul Health*. 2024;72:202198. doi: 10.1016/j.jep.2024.202198
42. Murray DM, Varnell SP, Blitstein JL. Design and analysis of group-randomized trials: a review of recent methodological developments. *Am J Public Health*. 2004;94:423–432. doi: 10.2105/ajph.94.3.423
43. Turner EL, Prague M, Gallis JA, Fan L, Murray DM. Review of recent methodological developments in group-randomized trials: part 2—analysis. *Am J Public Health*. 2017;107:1078–1086. doi: 10.2105/AJPH.2017.303707
44. Rotondi MA; Comprehensive R Archive Network. CRTSize: Sample size estimation functions for cluster randomized trials [computer software]. Accessed November 5, 2025. <https://CRAN.R-project.org/package=CRTSize>
45. Cody J. Sample size calculation using SAS®, R, and nQuery Software. In: Proceedings of the SAS Global Forum; Cary, NC: SAS Institute; 2020. Paper 4675.
46. Namit K, Allton C. SURVEYCLUSTER: Stata module to calculate sample size for cluster-based surveys to achieve desired precision [computer software]. Statistical Software Components S459499. Boston College Department of Economics. Accessed November 5, 2025. <https://ideas.repec.org/c/boc/bocode/s459499.html>
47. Carvajal SC, Baumler E, Harrist RB, Parcel GS. Multilevel models and unbiased tests for group-based interventions: examples from the safer choices study. *Multivariate Behav Res*. 2001;36:185–205. doi: 10.1207/S15327906MBR3602_03
48. Campbell MK, Piaggio G, Elbourne DR, Altman DG; CONSORT Group. CONSORT 2010 statement: extension to cluster randomised trials. *BMJ*. 2012;345:e5661. doi: 10.1136/bmj.e5661
49. Weijer C, Grimshaw JM, Eccles MP, McRae AD, White A, Brehaut JC, Taljaard M; Ottawa Ethics of Cluster Randomized Trials Consensus Group. The Ottawa statement on the ethical design and conduct of cluster randomized trials. *PLoS Med*. 2012;9:e1001346. doi: 10.1371/journal.pmed.1001346
50. Middleton S, McElduff P, Ward J, Grimshaw JM, Dale S, D'Este C, Drury P, Griffiths R, Cheung NW, Quinn C, et al; QASC Trialists Group. Implementation of evidence-based treatment protocols to manage fever, hyperglycaemia, and swallowing dysfunction in acute stroke (QASC): a cluster randomised controlled trial. *Lancet*. 2011;378:1699–1706. doi: 10.1016/S0140-6736(11)61485-2
51. Campbell MJ. Cluster randomized trials and stepped wedge trials. In: Ahrens W, Pigeot I, eds. *Handbook of Epidemiology*. Springer; 2019:1–38.
52. Proctor E, Silmere H, Raghavan R, Hovmand P, Aarons G, Bunger A, Griffey R, Hensley M. Outcomes for implementation research: conceptual distinctions, measurement challenges, and research agenda. *Adm Policy Ment Health*. 2011;38:65–76. doi: 10.1007/s10488-010-0319-7
53. Barker D, McElduff P, D'Este C, Campbell MJ. Stepped wedge cluster randomised trials: a review of the statistical methodology used and available. *BMC Med Res Methodol*. 2016;16:69. doi: 10.1186/s12874-016-0176-5
54. Hemming K, Haines TP, Chilton PJ, Girling AJ, Lilford RJ. The stepped wedge cluster randomised trial: rationale, design, analysis, and reporting. *BMJ*. 2015;350:h391. doi: 10.1136/bmj.h391
55. Baio G, Copas A, Ambler G, Hargreaves J, Beard E, Omar RZ. Sample size calculation for a stepped wedge trial. *Trials*. 2015;16:354. doi: 10.1186/s13063-015-0840-9
56. Caille A, Billot L, Kasza J. Practical and methodological challenges when conducting a cluster randomized trial: examples and recommendations. *J Epidemiol Popul Health*. 2024;72:202199. doi: 10.1016/j.jep.2024.202199
57. Hemming K, Taljaard M, McKenzie JE, Hooper R, Copas A, Thompson JA, Dixon-Woods M, Aldcroft A, Doussau A, Grayling M, et al. Reporting of stepped wedge cluster randomised trials: extension of the CONSORT 2010 statement with explanation and elaboration. *BMJ*. 2018;363:k1614. doi: 10.1136/bmj.k1614
58. Anderson CS, Summers D, Ouyang M, Sui Y, Johnson B, Billot L, Malavera A, Faigle R, Muñoz-Venturelli P, Day D, et al; OPTIMISTmain Investigators. Safety and efficacy of low-intensity versus standard monitoring following intravenous thrombolytic treatment in patients with acute ischaemic stroke (OPTIMISTmain): an international, pragmatic, stepped-wedge, cluster-randomised, controlled non-inferiority trial. *Lancet*. 2025;405:1909–1922. doi: 10.1016/S0140-6736(25)00549-5
59. Melo S, Bishop S. Translating healthcare research evidence into practice: the role of linked boundary objects. *Soc Sci Med*. 2020;246:112731. doi: 10.1016/j.socscimed.2019.112731
60. Curran GM, Landes SJ, McBain SA, Pyne JM, Smith JD, Fernandez ME, Chambers DA, Mittman BS. Reflections on 10 years of effectiveness-implementation hybrid studies. *Front Health Serv*. 2022;2:105349660. doi: 10.3389/frhs.2022.1053496
61. Curran GM, Bauer M, Mittman B, Pyne JM, Stetler C. Effectiveness-implementation hybrid designs: combining elements of clinical effectiveness and implementation research to enhance public health impact. *Med Care*. 2012;50:217–226. doi: 10.1097/MLR.0b013e3182408812
62. Handley MA, Murphy LD, Sherwin EB, Shade SB. Practical application of hybrid effectiveness-implementation studies for intervention research. *Int J Epidemiol*. 2025;54:dyaf039. doi: 10.1093/ije/dyaf039
63. Owen MA, Curran GM, Smith JD, Tedla Y, Cheng C, Spiegelman D. Power and sample size calculations for cluster randomized hybrid type 2 effectiveness-implementation studies. *StatMed*. 2025;44:e70015. doi: 10.1002/sim.70015
64. Owen MA; Comprehensive R Archive Network. crt2power: Designing cluster-randomized trials with two co-primary outcomes [computer software]. Version 1.1.0. Accessed November 5, 2025. <https://cran.r-project.org/web/packages/crt2power/>
65. Pinnock H, Barwick M, Carpenter C, Eldridge S, Grandes G, Griffiths CJ, Rycroft-Malone J, Meissner P, Murray E, Patel A, et al; StaRI Group. Standards for reporting implementation studies (StaRI) statement. *BMJ*. 2017;356:i6795. doi: 10.1136/bmj.i6795
66. Johnson AL, Ecker AH, Fletcher TL, Hundt N, Kauth MR, Martin LA, Curran GM, Cully JA. Increasing the impact of randomized controlled trials: an example of a hybrid effectiveness-implementation design in psychotherapy research. *Transl Behav Med*. 2020;10:629–636. doi: 10.1093/tbm/iby116
67. Trutschel D, Blatter C, Simon M, Holle D, Reuther S, Brunkert T. The unrecognized role of fidelity in effectiveness-implementation hybrid

trials: simulation study and guidance for implementation researchers. *BMC Med Res Methodol*. 2023;23:116. doi: 10.1186/s12874-023-01943-3

68. Fortney JC, Curran GM, Lyon AR, Check DK, Flum DR. Similarities and differences between pragmatic trials and hybrid effectiveness-implementation trials. *J Gen Intern Med*. 2024;39:1735–1743. doi: 10.1007/s11606-024-08747-1
69. Markle-Reid M, Fisher K, Walker KM, Beauchamp M, Cameron JI, Dayler D, Fleck R, Gafni A, Ganann R, Hajas K, et al. The stroke transitional care intervention for older adults with stroke and multimorbidity: a multisite pragmatic randomized controlled trial. *BMC Geriatr*. 2023;23:687. doi: 10.1186/s12877-023-04403-1
70. Markle-Reid M, Fisher K, Walker KM, Cameron JI, Dayler D, Fleck R, Gafni A, Ganann R, Hajas K, Koetsier B, et al. Implementation of the virtual transitional care stroke intervention for older adults with stroke and multimorbidity: a qualitative descriptive study. *J Multimorb Comorb*. 2025;15:26335565251323748. doi: 10.1177/26335565251323748
71. Gamerman V, Cai T, Elsässer A. Pragmatic randomized clinical trials: best practices and statistical guidance. *Health Serv Outcomes Res Methodol*. 2019;19:23–35. doi: 10.1007/s10742-018-0192-5
72. Fontaine G, Vinette B, Weight C, Maheu-Cadotte MA, Lavalée A, Deschênes MF, Lapierre A, Castiglione SA, Chicoine G, Rouleau G, et al. Effects of implementation strategies on nursing practice and patient outcomes: a comprehensive systematic review and meta-analysis. *Implement Sci*. 2024;19:68. doi: 10.1186/s13012-024-01398-0
73. Loudon K, Treweek S, Sullivan F, Donnan P, Thorpe KE, Zwarenstein M. The PRECIS-2 tool: designing trials that are fit for purpose. *BMJ*. 2015;350:h2147. doi: 10.1136/bmj.h2147
74. Kitson A, Harvey G, McCormack B. Enabling the implementation of evidence based practice: a conceptual framework. *Qual Health Care*. 1998;7:149–158. doi: 10.1136/qshc.7.3.149
75. Grant A, Treweek S, Dreischulte T, Foy R, Guthrie B. Process evaluations for cluster-randomised trials of complex interventions: a proposed framework for design and reporting. *Trials*. 2013;14:15. doi: 10.1186/1745-6215-14-15
76. Birken SA, Rohweder CL, Powell BJ, Shea CM, Scott J, Leeman J, Grewe ME, Kirk MA, Damschroder L, Aldridge WA II, et al. T-CaST: an implementation theory comparison and selection tool. *Implement Sci*. 2018;13:143. doi: 10.1186/s13012-018-0836-4
77. Fontaine G, Mooney M, Porat-Dahlerbruch J, Cahir K, Ellen M, Spinewine A, Taylor N, Laritz R, Bourbeau-Allard E, Grimshaw JM. Advancing the selection of implementation science theories, models, and frameworks: a scoping review and the development of the SELECT-IT meta-framework. *Implement Sci*. 2025;20:24. doi: 10.1186/s13012-025-01436-5
78. Weight C, Vinette B, Laritz R, Mooney M, Castiglione SA, Maheu-Cadotte MA, Argiropoulos N, Konnyu K, Cassidy CE, Collins SE, et al. How well are implementation strategies and target healthcare professional behaviors reported? A secondary analysis of 204 implementation trials using the TIDieR checklist and AACTT framework. *Implement Sci*. 2025;20:28. doi: 10.1186/s13012-025-01442-7
79. Peterson HB, Dube Q, Lawn JE, Haidar J, Bagenal J, Horton R; Lancet Commission on Evidence-Based Implementation in Global Health. Achieving justice in implementation: the Lancet Commission on Evidence-Based Implementation in Global Health. *Lancet*. 2023;402:168–170. doi: 10.1016/S0140-6736(23)00870-X
80. Middleton S, Levi C, Ward J, Grimshaw JM, McElduff P, D'Este C, Drury P, Griffiths R, Cheung NW, Quinn C, et al. Fever, hyperglycaemia and swallowing dysfunction management in acute stroke: a cluster randomised controlled trial of knowledge transfer. *Implement Sci*. 2009;4:16. doi: 10.1186/1748-5908-4-16
81. Gallo A, Weijer C, White A, Grimshaw JM, Taljaard M, Brehaut JC, McRae AD, Eccles MP. What is the role and authority of gatekeepers in cluster randomized trials in health research? *Trials*. 2012;13:116. doi: 10.1186/1745-6215-13-116
82. Flodgren G, O'Brien MA, Parmelli E, Grimshaw JM. Local opinion leaders: effects on professional practice and healthcare outcomes. *Cochrane Database Syst Rev*. 2019;6:CD000125. doi: 10.1002/14651858.CD000125.pub5
83. Grol R, Wensing M, Eccles M. Implementation of changes in practice. In: Grol R, Wensing M, Eccles M, eds. *Improving Patient Care: the Implementation of Change in Clinical Practice*. Elsevier; 2005:41–64.
84. Hamilton S, McLaren S, Mulhall A. Assessing organisational readiness for change: use of diagnostic analysis prior to the implementation of a multidisciplinary assessment for acute stroke care. *Implement Sci*. 2007;2:21. doi: 10.1186/1748-5908-2-21
85. Grol R, Grimshaw J. Evidence-based implementation of evidence-based medicine. *Jt Comm J Qual Improv*. 1999;25:503–513. doi: 10.1016/s1070-3241(16)30464-3
86. Forsetlund L, Bjørndal A, Rashidian A, Jamtvedt G, O'Brien MA, Wolf F, Davis D, Odgaard-Jensen J, Oxman AD. Continuing education meetings and workshops: effects on professional practice and health care outcomes. *Cochrane Database Syst Rev*. 2009;2009:CD003030. doi: 10.1002/14651858.CD003030.pub2
87. O'Brien MA, Rogers S, Jamtvedt G, Oxman AD, Odgaard-Jensen J, Kristoffersen DT, Forsetlund L, Bainbridge D, Freemantle N, Davis DA, et al. Educational outreach visits: effects on professional practice and health care outcomes. *Cochrane Database Syst Rev*. 2007;2007:CD000409. doi: 10.1002/14651858.CD000409.pub2
88. Grimshaw J, Thomas R, MacLennan G, Fraser C, Ramsay C, Vale L, Whitty P, Eccles M, Matowe L, Shirran L, et al. Effectiveness and efficiency of guideline dissemination and implementation strategies. *Health Technol Assess*. 2004;8:iii–xii, 1–72. doi: 10.3310/hta8060
89. Edmondson AC, Bohmer RM, Pisano GP. Disrupted routines: team learning and new technology implementation in hospitals. *Administr Sci Quarter*. 2001;46:685–716. doi: 10.2307/3094828
90. Green PL. Improving clinical effectiveness in an integrated care delivery system. *J Health Qual*. 1998;20:4–8; quiz 9, 48. doi: 10.1111/j.1945-1474.1998.tb00289.x
91. Sengul A, Escobar E, Flores JR, Kwok M, Kono S, Guyatt G, Jackevicius CA. Non-inferiority trials: a systematic review on methodological quality and reporting standards. *J Gen Intern Med*. 2024;39:2522–2530. doi: 10.1007/s11606-024-08890-9
92. Billot L, Bompont S, Ouyang M, Xu L, Wang X, Malavera A, Johnson B, Summers D, Khatri P, Muñoz-Venturelli P, et al. Statistical analysis plan for the optimal post rTpa-Iv monitoring in ischemic stroke trial (OPTIMIST) [published online December 18, 2024]. *medRxiv*. doi: 10.1101/2024.12.17.24319094
93. Ouyang M, González F, Montalbano M, Nguyen TN, Malavera A, Johnson B, Summers D, Khatri P, Muñoz-Venturelli P, Day D, et al. Implementation of low-intensity thrombolysis monitoring care in routine practice: process evaluation of the optimal post rTpa-IV monitoring in acute ischemic stroke study in the USA. *Cerebrovasc Dis*. 2024;54:96–104. doi: 10.1159/000538136
94. Torrealba-Acosta G, Garg R, Mandava P. What OPTIMISTmain teaches about margins, imputation, and interpretation. *Stroke Clinician*. 2025;2. doi: 10.59236/sc.v2i4.102
95. Murray E, Treweek S, Pope C, MacFarlane A, Ballini L, Dowrick C, Finch T, Kennedy A, Mair F, O'Donnell C, et al. Normalisation process theory: a framework for developing, evaluating and implementing complex interventions. *BMC Med*. 2010;8:63. doi: 10.1186/1741-7015-8-63